SYNTHETIC DEMONSTRATION DATA. NOT VALID FOR REAL MACHINE OPERATION.

Quality Alert: Titanium Grade 5 Chatter & Feed Rate Restriction

Page 1: Section 1: Problem Description & Defect Root Cause

QUALITY ALERT QA-CNC-2024-03 | Effective Date: 2024-03-10 | Severity: HIGH PRIORITY

Defect Summary: Surface chatter marks exceeding Ra 1.6 ?m and micro-chipping on Ti-6Al-4V turbine bracket flanges produced on DEMO-CNC-01. Root cause analysis determined that high cutting forces at 1,500 mm/min feed rate induced harmonic resonance in workpiece fixtures.

SYNTHETIC DEMONSTRATION DATA. NOT VALID FOR REAL MACHINE OPERATION.

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Page 2: Section 2: Mandatory Operational Restriction

MANDATORY OVERRIDE FOR ALL DEMO-CNC-01 OPERATORS:
Maximum allowable feed rate for face milling Titanium Grade 5 is strictly capped at 1,200 mm/min (Reduced from general catalog limit of 1,500 mm/min).
Do NOT exceed 1,200 mm/min feed rate under any circumstance until fixture redesign (ECN-892) is released.

Tooling Requirement: Use 4-flute solid carbide endmill with 0.5 mm corner radius and minimum 8% water-soluble coolant flow.

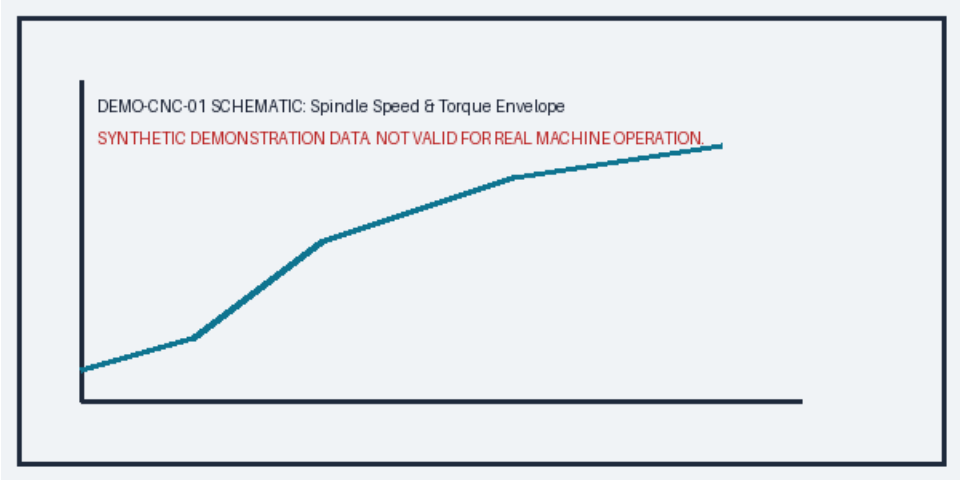


Figure 1: QA-CNC-2024-03 Operational Limits Diagram